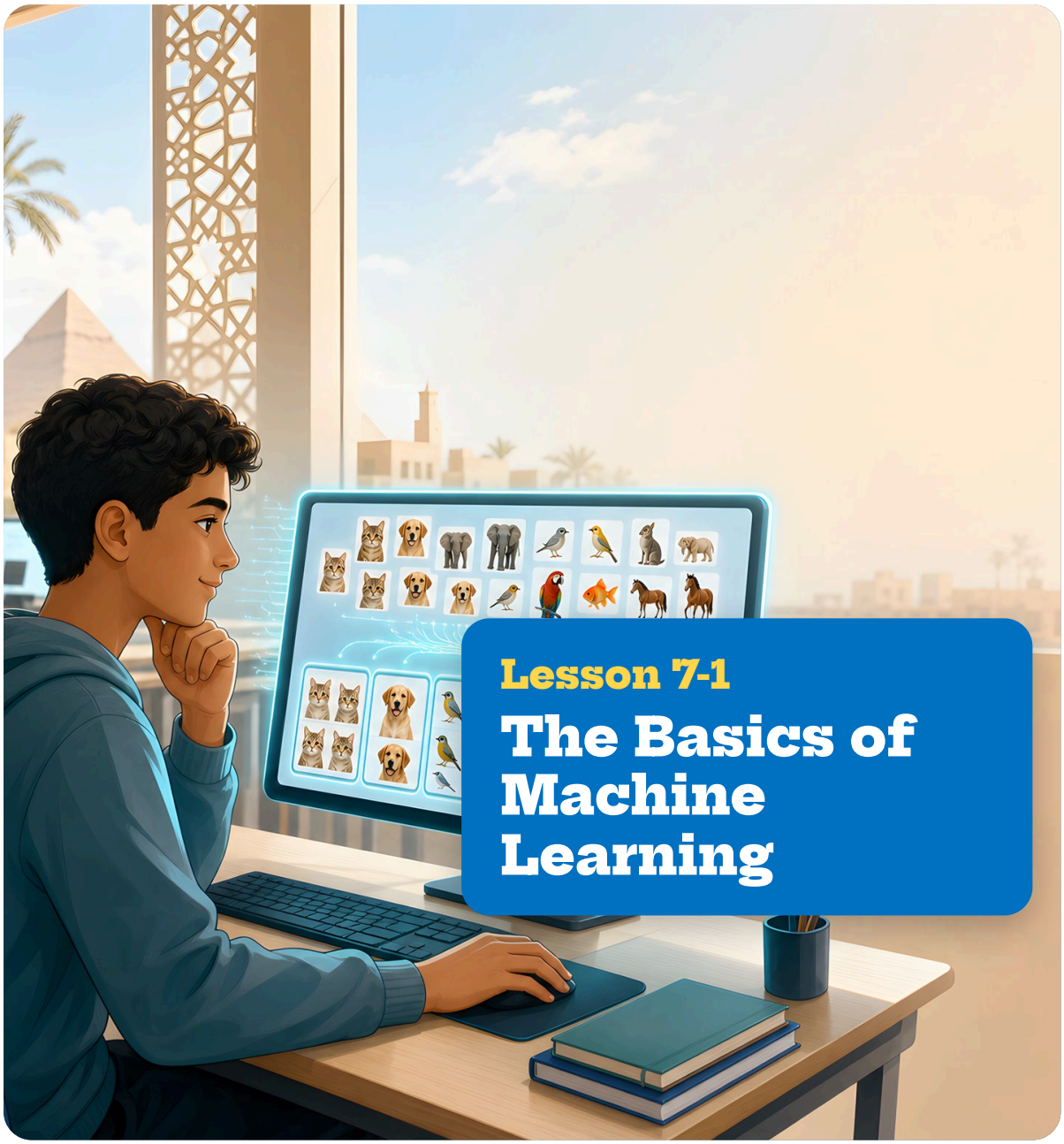


Lesson 7-1

The Basics of Machine Learning

Learning Objectives

- Explain the idea of machine learning and distinguish supervised, unsupervised, and reinforcement learning.
- Describe classification, clustering, and reinforcement learning with examples, and explain the limits caused by data bias and the black-box problem.

How do you teach a computer to tell a cat from a dog when you cannot write down a rule for it? You do not write the rule — you show it thousands of examples and let it find the rule itself. That is machine learning, the engine behind most of today's AI. It learns patterns from data to make judgments on new data. Yet it is not magic: if the examples are biased, the judgments are biased, and its reasoning can be a black box even to its makers. Machine learning is therefore best used to support human judgment, not to replace it.

? Guiding question: How does machine learning solve problems differently from ordinary programming, and what are its main types and limits?

Point!

1. The Idea of Machine Learning

(1) **(Machine learning)**…a technology that learns patterns and regularities from large amounts of data so that judgments and predictions can be made on new data. It is the most common method used to make AI work.

(2) Conventional programming and machine learning differ greatly in how they solve problems.

Approach	What humans provide	What the computer does
Conventional programming	Decision rules (procedures) and input data	Produces answers by following the rules
Machine learning	Large amounts of data (with correct answers, where available)	Finds the decision rules from the data on its own

(3) Machine learning is used for problems where it is difficult for humans to express the rules in words. *e.g.,* judging whether a photo shows a dog or a cat; distinguishing whether an email is spam; transcribing audio into text

(4) For such problems, the rules are too complex for humans to write out completely, so the approach of letting the computer learn from examples (data) becomes effective.

(5) However, machine learning is not a panacea. Because the quality and bias of the training data affect the results, biased data leads to biased judgments (bias). There is also the black-box problem, in which the basis for the model's judgments becomes hard for humans to see. Given these limitations, it is important to use machine learning not as a replacement for human judgment, but as a tool that supports human judgment.

2. Three Types of Learning Methods

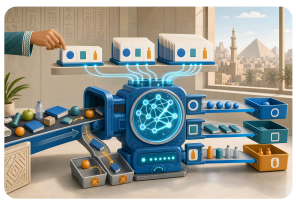
(1) Machine learning is broadly divided into the following three types, depending on the form of data given and the method of learning.

Key Fact:
Machine learning finds the rules from data instead of being given them. It comes in three forms — supervised (learn from labeled examples), unsupervised (find hidden groups), and reinforcement (learn from rewards) — but its results depend on the data, so bias and the black-box problem make it a support for human judgment, not a replacement.

Key Concepts:

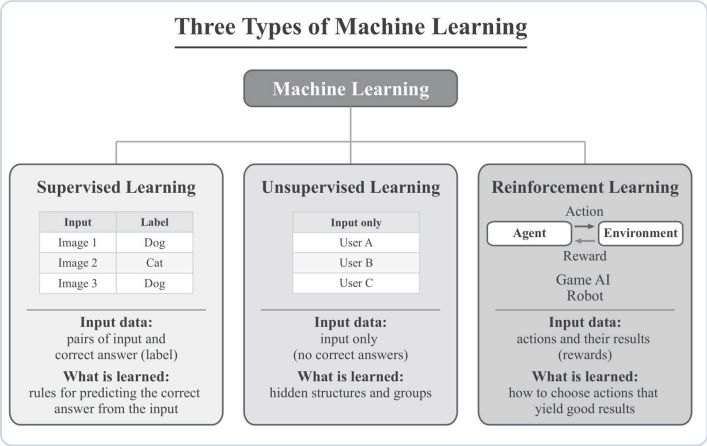
- machine learning
- supervised / unsupervised / reinforcement
- classification, feature
- clustering
- bias, black-box problem

KEY TERMS
machine learning — learning rules from data to judge new data.
bias — skewed data leading to skewed judgments.
black-box problem — the basis of a judgment is hard to see.



Machine learning improves from more examples.

Learning method	Data given	What is learned
(Supervised learning)	Pairs of input and the correct answer (label)	Rules for predicting the correct answer from the input
(Unsupervised learning)	Input only (no correct answers)	Hidden structure or groups in the data
(Reinforcement learning)	Actions and their results (rewards)	How to choose actions to obtain better results



The three types of machine learning and the data each uses

(2) Which learning method to use is determined by the type of data at hand and the nature of the problem to be solved.

Situation	Suitable learning method
Past data has correct answers, and you want to predict the correct answer for new data	Supervised learning
Data is available but without correct answers, and you want to find tendencies or groups in the data	Unsupervised learning
You want the system to learn how to choose actions for better results through trial and error	Reinforcement learning

KEY TERMS
unsupervised learning — finds groups with no labels.
reinforcement learning — learns actions from rewards.
label — the known correct answer on training data.

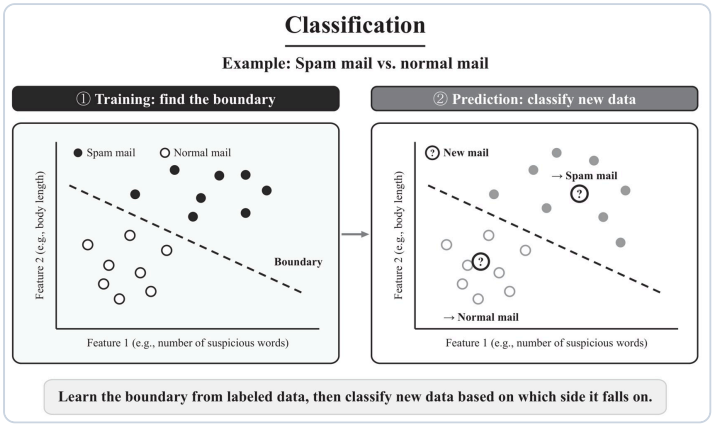
3. Classification (A Representative Example of Supervised Learning)

(1) **(Classification)**…a method of predicting which type (class) the input data belongs to. It is a representative example of supervised learning. *e.g.,* sorting emails into “spam” or “normal”; sorting medical images into “abnormal” or “normal”; classifying animals in photos as “dog,” “cat,” or “bird”

(2) In classification, large amounts of data with correct answers (labels) attached are given in advance for learning. After training is complete, the model can predict which class new data without known labels belongs to.

KEY TERMS
supervised learning — learns from labeled examples.
classification — predict which class an input belongs to.
feature — a clue in the data useful for prediction.

Stage	What is done
(Training)	Find rules for classification from pairs of input data and correct answers (labels)
(Prediction)	Use the learned rules to determine which class new data belongs to



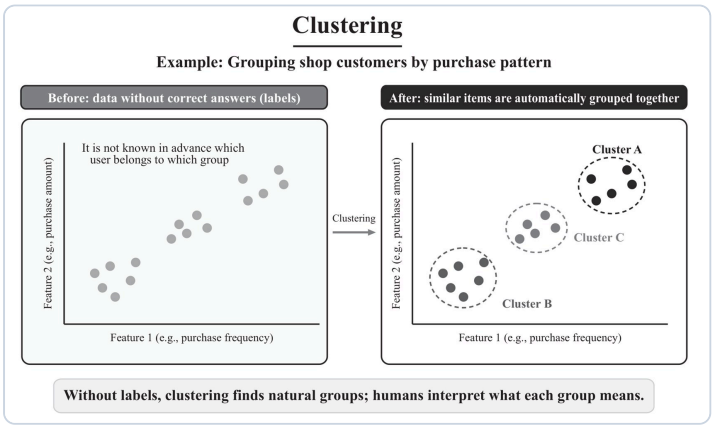
Classification: learn a boundary from labeled data, then classify new data

(3) Information in the training data that serves as a clue useful for prediction is called a **(feature)**. *e.g.,* In spam classification, the types of words contained in the email, the sender's address, and the length of the body are features.

(4) The choice of features greatly affects the accuracy of classification. Choosing features that match the purpose is important for building a good classification model.

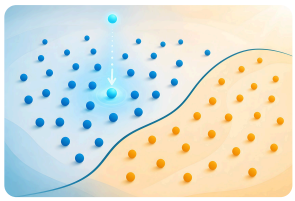
4. Clustering (A Representative Example of Unsupervised Learning)

(1) **(Clustering)**…a method that groups similar items together into groups (clusters) within data. It is a representative example of unsupervised learning. *e.g.,* grouping users of an online shop by similar purchase tendencies; grouping users with similar listening patterns on a music streaming service for recommendations; automatically grouping news articles with similar content



Clustering: without labels, similar items are automatically grouped

(2) The difference between clustering and classification lies in whether the correct answers (labels) are given.



Classification draws a boundary between labeled groups.

PAUSE & THINK

Labels given or not given — which one word separates classification from clustering?

Method	Correct answers (labels)	Purpose
Classification	Given in advance	Predict which class new data belongs to
Clustering	Not given	Find natural groups within the data

(3) In clustering, the computer itself finds "what is similar." Even when the names or number of groups are not known in advance, grouping can be done automatically based on the features of the data.

(4) Clustering is used when one wants to grasp the overall picture of the data or when one wants to look at the data from a new perspective. Humans interpret the results of grouping and use them in marketing, product development, and so on.

5. Reinforcement Learning

(1) **(Reinforcement learning)**…a method in which actions are chosen in a certain situation, and how to choose better actions is learned based on the reward obtained as a result. *e.g.,* learning how to obtain a high score in a game through trial and error; a robot learning to walk without falling by trying movements; a self-driving car learning safe driving operations

(2) In reinforcement learning, learning is accomplished by repeatedly trying combinations of actions and results.

Element	Description
(State)	The situation in which the agent (the learning side) is placed
(Action)	The choices that can be selected in that situation
(Reward)	The evaluation given as a result of the action (a numerical value indicating goodness or badness)

(3) What distinguishes reinforcement learning from the other two learning types is that it learns from the consequences of its own actions. Unlike supervised learning, where there is a correspondence of "this input has this correct answer," reinforcement learning gradually learns better choices from the rewards obtained for the actions it tries.

(4) Reinforcement learning is suited to problems where the correct answer is not known in advance, but the goodness of the result can be judged. It is often used in combination with environments where trial and error can be repeated (simulators or games).

KEY TERMS

clustering — group similar items with no labels.
reinforcement learning — learn actions from rewards.
reward — a value scoring how good an action was.



Reinforcement learning learns by reward and penalty.

PAUSE & THINK

Learning from rewards through trial and error — which of the three types is that?

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A Machine learning is a technology that learns patterns and regularities from large amounts of data so that judgments and predictions can be made on new data.

B In supervised learning, the model learns from data without correct answers (labels).

C Clustering is a method that groups data according to labels given in advance.

D Reinforcement learning is a method in which how to choose better actions is learned based on the reward obtained as a result of an action.

(2) Match each description (a–c) with the most appropriate term from the choices below (A–C).

a A learning method that learns rules for predicting correct answers for inputs from pairs of input and correct answer

b A method that finds groups of similar items from data without correct answers

c Information in the input data that serves as a clue for prediction in classification

[Choices] **A** Feature **B** Supervised learning **C** Clustering

✓ Solution

(1)

A Correct definition of machine learning. Therefore, ○.

B Data without correct answers (labels) is unsupervised learning. In supervised learning, correct answers are given. Therefore, ×.

C Clustering is a method that finds natural groups from data without given labels. Therefore, ×.

D Correct description of reinforcement learning. Therefore, ○.

(2)

a: Learning from pairs of input and correct answer is supervised learning. **B**

b: Finding groups from unlabeled data is clustering. **C**

c: Information that serves as a clue for prediction is a feature. **A**

Try

1 Answer the following questions.

1. What is the term for the technology that learns patterns and regularities from large amounts of data so that judgments and predictions can be made on new data?
2. In machine learning, what is the term for the information in the input data that serves as a clue useful for prediction?
3. What is the term for the representative method of unsupervised learning that groups similar items together?
4. In reinforcement learning, what is the term for the numerical value that represents the evaluation given as a result of an action?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that is NOT an appropriate situation in which machine learning is used.

A Distinguishing the type of animal in a photograph.

B Transcribing audio into text.

C Calculating the sum of two integers.

D Distinguishing whether an email is spam.

(2) For each of purposes (a–c), choose the most appropriate learning method from the choices below (A–C).

a Predict, from past purchase history data and whether or not a purchase was made, whether a new customer will buy the product.

b Find groups of customers with similar purchase tendencies from customer purchase data.

c Have a robot walk and learn movements that allow it to advance without falling through trial and error.

[Choices] **A** Unsupervised learning **B** Supervised learning **C** Reinforcement learning

EXAM-STYLE QUESTION

Explain why machine learning should support human judgment rather than replace it. **[6] Refer to:** bias in data and the black-box problem.

PAUSE & THINK

Which task has a simple rule a human can already write — so needs no learning?

(3) From the following options (A–D), choose the one that most appropriately describes the difference between classification and clustering.

A Classification divides data into a small number of groups, and clustering divides data into many groups.

B Classification handles only numerical data, and clustering handles only text data.

C Classification learns from data with correct answers (labels), and clustering learns from data without correct answers.

D Classification is unsupervised learning, and clustering is supervised learning.

(4) From the following options (A–D), choose the one that is NOT an appropriate point of caution when using machine learning.

A If the training data is biased, the model's judgments may also be biased.

B The basis for a machine learning model's judgments may be hard to see, so for important judgments human checking is needed.

C Machine learning is a panacea and can make more accurate judgments than humans on any problem.

D It is important to use machine learning as a tool that supports human judgment.

⚙ Think as an Engineer — Investigate & Decide

Your school has years of data and wants help with two tasks: (i) flagging students who may need extra support early, and (ii) discovering natural groups of study habits it did not know about. You advise on how to use machine learning responsibly.

☐ Let the model decide alone **☐ Choose the right method, and keep a human in charge**

1 · Identify the method. For task (i) and task (ii), name which learning type fits (supervised, unsupervised, or reinforcement). For each, name one piece of data the school would need to collect, and one kind of student that data might miss.

2 · Analyze the risks. Name one way the training data for task (i) could be biased (for example, if it only includes students who already asked for help), and say what harm the black-box problem could cause if a teacher acts on a flag without understanding it — a judgment the lesson's tables do not make for you.

3 · Decide the role of the human. State which decisions you would keep a human in charge of, and which you would let the model assist with, and why. **In pairs,** compare plans and agree on the one safeguard you would never remove.

Give evidence for your answer.

KEY TERMS

label — the known correct answer attached to data.

human-in-the-loop — a person checks or approves the model's output.

Stuck? Start from the data you have: are there correct answers to learn from (supervised) or none (unsupervised)? Then ask, for each decision, what it would cost to be wrong — the higher that cost, the more a human must stay in charge.

🌍 In a New Context

A bank wants software that reads each loan application and flags the ones that look risky for a human officer to review. Name which learning type this is, name one feature the model might use, and explain why the officer must stay in the loop rather than letting the model decide alone.

❓ Lesson Question — Answered

The lesson asked how machine learning differs from ordinary programming and what its main types and limits are. In conventional programming, humans provide the rules and the computer follows them; in machine learning, humans provide data and examples, and the computer finds the rules itself — which is why it suits problems whose rules are too complex to write, like recognizing images or speech. It comes in three types: supervised learning (labeled examples, whose representative task is classification, using features to predict a class), unsupervised learning (no labels, whose representative task is clustering, finding natural groups), and reinforcement learning (learning from rewards through trial and error, described by state, action, and reward). Its limits matter as much as its power: results depend on the quality and bias of the data, so biased data gives biased judgments, and the black-box problem hides the basis of a decision. For these reasons machine learning is best used to support human judgment, not to replace it.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

Machine learning is a technology that learns patterns and regularities from large amounts of data so that judgments and predictions can be made on new data. Machine learning is broadly divided into three types depending on the form of data given and the method of learning. There is (a) learning, which learns from pairs of input and correct answer; (b) learning, which finds structure or groups from data without correct answers; and (c) learning, which learns how to choose better actions based on the (d) obtained as a result of an action. In (e), a representative example of supervised learning, the choice of (f), which serves as a clue for prediction, greatly affects the result. In (g), a representative example of unsupervised learning, the computer itself finds "what is similar" and discovers natural groups.

1. Fill in the blanks (a)–(g).

3 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes the difference between conventional programming and machine learning.

A In conventional programming, humans provide the decision rules; in machine learning, the computer finds the decision rules from data on its own.

B Conventional programming requires large amounts of data, and machine learning requires only rules.

C Conventional programming is included in AI, but machine learning is not included in AI.

D Conventional programming can handle new problems, but machine learning can handle only fixed problems.

(2) For each of situations (a–c), choose the most appropriate term from the choices below (A–C).

a There are large amounts of news article texts and their categories (politics, economics, sports, etc.), and you want to automatically determine the category of new articles.

b You have records of products purchased by users of an online shop and want to find groups of users with similar purchasing tendencies.

c You want a computer to play a game and learn the operations that lead to high scores through trial and error.

[Choices] **A** Reinforcement learning **B** Clustering **C** Classification

REFLECT & CHALLENGE

Reflect: Where have you likely met machine learning today (recommendations, photo tagging, voice input)? Which type do you think it was?
Challenge: A face-recognition model was trained mostly on one group of people and works poorly on others. Explain how biased training data caused this, and state one change to the data that would reduce the bias.

Key takeaway

Remember:
Machine learning finds rules from data: supervised (labeled → classification with features), unsupervised (unlabeled → clustering), reinforcement (rewards). Mind its limits — bias and the black-box problem — and keep it supporting human judgment.

(3) From the following options (A–D), choose the one that is NOT an appropriate description of the characteristics and limitations of machine learning.

A Machine learning is used for problems where it is difficult for humans to express the rules in words.

B Machine learning models are affected in their results by the quality and bias of the training data.

C A machine learning model's basis for judgment is always shown to humans in an easy-to-understand way.

D It is important to use machine learning not as a replacement for human judgment, but as a tool that supports human judgment.